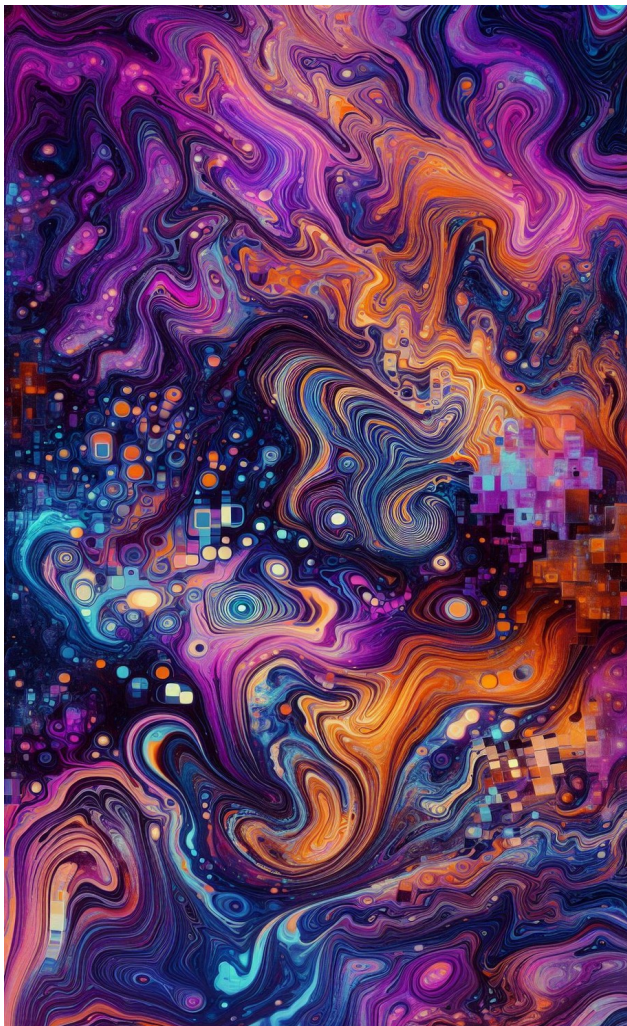




GLOBAL FIELD MODULATION IN SEQUENCE PROCESSING:

*Impact on synchrony, rank efficiency, interference and
autocorrelation*

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*As light moves like a wave.
As water thrashes the shore.
It can't be.
That only we,
move through space.
Like a wave.
Like a wave.
Ought to serve.
It observes.*

Abstract

This thesis explores neural networks inspired by biological Local Field Potentials (LFPs), abstracted as slowly evolving Global Fields (GF) that modulate the activity of low-parameter neural populations. Synthesising the Communication-through-Coherence principle with spectral analysis, the work evaluates whether diffuse modulation mitigates representational collapse. Unlike standard networks, GF Neural Networks introduce signals derived from population activity that influence dynamics over time.

The work evaluates whether such modulation affects learning dynamics, representational structure, and interference by means of Sobol sensitivity sweeps and comparing performance against baseline and "probing" architectures.

Benchmarking both independently recurrent neural networks(indRNN) and standard RNN . Evaluation datasets include sequential and permuted MNIST (sMNIST and psMNIST), the Adding and Copying tasks, and the sequential CIFAR-10 (sCIFAR-10) dataset. These test for long-range dependency and credit assignment, temporal buffering and gradient stability, representational interference and forgetting, and dimensional collapse.

Key findings reveal the GF acts as a task-dependent regulator. Sobol analysis identifies a H_{inertia} bifurcation threshold, paralleling "edge-of-chaos" dynamics, where networks achieve productive metastability. In sequential CIFAR-10(sCIFAR-10), the GF acts as an integrator, yielding a 4.6-fold accuracy improvement and superior weight efficiency relative to comparably-sized recurrent baselines. Conversely, it disrupts tasks requiring high unit independence, like the Adding Task. Ultimately, the GF enables coordination-without-collapse, fostering functional integration in architectures lacking inherent cross-unit communication.

1. Introduction

Biological neural systems exhibit population-level signals such as Local Field Potentials (LFPs), reflecting collective activity that evolves on slower timescales than individual spikes (Belitski et al., 2008; Esghaei et al., 2018). Standard artificial architectures, including feed-forward, convolutional, recurrent, and attention-based models, have no equivalent mechanism. Units are driven by local inputs and weighted connections with no explicit population-level state that evolves over time. This creates a structural gap: the network has no capacity to sense or regulate its own collective dynamical state during learning.

This absence may contribute to a well-documented failure mode. Prior work has shown that many architectures exhibit representational collapse during training, where activity becomes increasingly low-rank or redundant, reducing effective capacity (Papayan et al., 2020; Roth et al., 2024). One possibility is that purely local learning dynamics lack the global regulatory signals needed to stabilise population-level structure. While current approaches to long-range sequence modelling, from Transformers (Vaswani et al., 2017) to Structured State Spaces (Gu et al., 2022) and selective variants like Mamba (Gu and Dao, 2024), address temporal dependencies through various mechanisms, all treat units as independent streams with no cross-unit coordination. The theoretical and empirical implications of this are examined in Section 2.

This study introduces a GF, a shared, slowly evolving signal derived from population activity, to test whether weak population-level coupling can act as a regulatory mechanism. Two objectives structure the investigation. Condition 1 treats the GF as an internal probe: it is computed as the mean population activity within a layer but decoupled from the forward pass. This condition establishes an observational baseline, tracking how population-level signals evolve during training without influencing it. Condition 2 treats the GF as a modulator: it is implemented as a leaky integrator that modulates neuronal pre-activations. This condition introduces global temporal coupling absent from standard designs and is compared against both the probe and a passive baseline to isolate the causal effect of field modulation.

To dissociate representational structure from task optimisation, the thesis evaluates population-level dynamics using epoch-wise measures of Effective Rank(ER) (Roy & Vetterli, 2007), Synchrony (Uhlhaas et al., 2009), Entropy (Schreiber, 2000; Keller et al., 2019), interference, and Temporal autocorrelation (Shi et al., 2023). These metrics quantify representational stability. The 'Reversal Memory Check' serves as a proxy for temporal sensitivity: if a network's hidden states are anchored by GF they should exhibit a degree of 'Temporal Invariance', the ability to maintain performance even when the input order is inverted.

1.1 Research Questions (RQ)

- **RQ1:** Does GF modulation correlate with stabilised ER and synchrony under temporal stress compared to passive baselines?
- **RQ2:** Does GF modulation actively reduce representational interference and facilitate "Active Maintenance" during sequence delays?

1.2 Hypotheses

Based on these questions, the study tests two primary hypotheses regarding the regulatory role of the GF:

Hypothesis 1: Representational Rigidity (The Stability Prediction for RQ1)

Statement: Active GF modulation acts as a structural scaffold that stabilizes the internal rank of the network, preventing the dimensional collapse typically observed during high-noise events or gradient instability. Under conditions of temporal stress, the GF will preserve a higher functional capacity than standard recurrent updates.

- **Success Criterion (Stability):** The variance of the ER (σ_{ER2}) throughout training must be significantly lower ($p < 0.05$) in the active condition than in the passive baseline.
- **Success Criterion (Temporal Robustness):** The retention ratio must remain significantly higher in the active condition, proving that the stabilised rank translates to a functional ability to handle non-standard sequence orientations.

Hypothesis 2: Non-Pathological Coordination (The Bifurcation Proof)

Statement: There exists a "Goldilocks" regime of modulation strength where global coordination occurs without falling into pathological entrainment or redundancy.

- **Success Criterion (The Bifurcation Point):** A Sobol sensitivity sweep must identify a critical transition point where increasing modulation strength triggers a phase shift, marked by a sharp increase in interference (> 0.7) and a corresponding decrease in ER, leading to a collapse in accuracy.
- **Success Criterion (Validation of Segregation):** Success is defined by identifying a parameter regime where Synchrony is significantly elevated above the probe baseline while interference remains low (< 0.4), empirically proving coordinated temporal binding without a loss of functional segregation.

2. Background and Theory

The following sections develop the theoretical foundation for the GF mechanism by tracing four converging lines of evidence: that biological neural populations use slow, rhythmic signals to coordinate information flow through coherence rather than rewiring (Section 2.1); that artificial sequence architectures, despite advances in long-range dependency modelling, eliminate this cross-unit coordination in pursuit of efficiency (Section 2.2). This absence leaves networks without a regulatory mechanism to distinguish beneficial compression from pathological representational collapse (Section 2.3); and that slowly evolving aggregate signals are well-suited to provide such regulation, as biological populations maintain stable collective geometry through drift by anchoring unit dynamics to a shared reference (Section 2.4). Together these threads motivate a GF: a slowly evolving, rhythmically modulated feedback signal derived from population activity that closes the regulatory loop absent from current architectures. A schematic mapping of these lineages is provided in Figure 1.

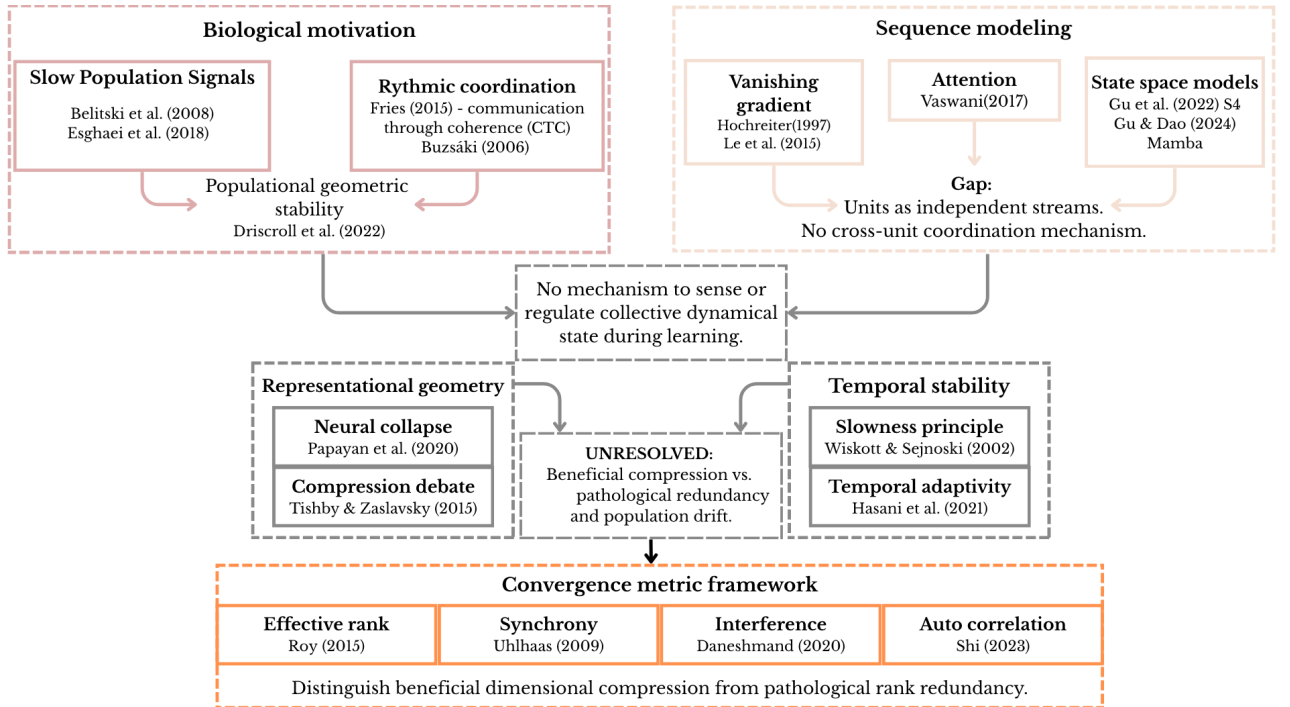


Figure 1: **Conceptual framework** integrating biological motivation and sequence modelling to address representational stability

2.1 Population-Level Coordination in Biological and Artificial Systems

Biological neural populations exhibit collective signals, such as Local Field Potentials (LFPs), that aggregate activity across ensembles and evolve on slower timescales than individual spikes (Belitski et al., 2008; Esghaei et al., 2018). These signals are not merely epiphenomenal. Evidence suggests they shape excitability and modulate information routing between regions. Fries (2015) formalises this in the Communication-through-Coherence (CTC) hypothesis, proposing that rhythmic alignment between populations gates communication without requiring rewiring. However, CTC remains debated; critics argue correlations between coherence and communication may reflect shared input rather than a direct mechanistic role (Chalk et al., 2016). Despite this, the core observation, that population-level signals on slower timescales structure the temporal windows of information flow, has not been refuted empirically.

It is worth distinguishing the GF from existing slow-scale signals in standard training pipelines. Batch normalisation running statistics, Adam exponential moving averages, and momentum terms operate on the optimisation trajectory rather than representational dynamics. They regulate how weights change, not how activations coordinate during forward processing. The GF closes a feedback loop these mechanisms leave open: it is derived from population activity during the forward pass and returned to that population as a modulatory signal in real time.

Standard artificial architectures do not implement an equivalent mechanism. Feed-forward, convolutional, and recurrent networks operate through local unit-to-unit interactions mediated by fixed weights, with no explicit population-level state evolving over time. Even architectures for long-range dependencies, such as Structured State Spaces (Gu et al., 2022) and Mamba (Gu and Dao, 2024), process channels as independent 1D controllers with no cross-unit coordination. This represents a structural gap: these models lack a mechanism to sense or regulate their own collective dynamical state during processing.

To address this gap, this study introduces a GF formalized as a leaky integrator over the layer's population activity:

$$GF_t = (1 - \lambda)GF_{t-1} + \lambda \cdot h_{t-1}$$

M1: GF as a leaky integrator Where λ controls the rate of temporal decay and h_{t-1} is the mean hidden state.

This single equation is the basis against which the following theoretical framework is evaluated. The CTC framework thus provides a theoretical framework for the core architectural choice of this study: that rhythmic, population-level signals operating on slower timescales can coordinate information flow without requiring additional synaptic connections.

2.2 Long-Range Temporal Dependencies in Sequence Models

Sequence modelling tasks requiring long-range credit assignment, such as (permuted) sequential MNIST, – CIFAR-10((p)sMNIST;sCIFAR-10)(LeCun et al., 1998;Krizhevsky 2009), and the adding problem(Hochreiter and Schmidhuer, 1997), expose fundamental limitations in RNNs. Hochreiter and Schmidhuber (1997) demonstrated that RNNs suffer from vanishing gradients across long sequences and Le et al. (2015) showed that, maintaining stable hidden states across hundreds of time steps remains difficult. LSTMs partially address this through gating, but struggle on tasks exceeding several hundred steps without augmentation.

Transformer architectures achieve global context through self-attention, but at quadratic complexity $O(L^2)$ relative to sequence length, making them computationally expensive (Vaswani et al., 2017; Tay et al., 2022). Structured State Space models (S4; Gu et al., 2022) and selective variants like Mamba (Gu and Dao, 2024), Mamba-2 (Dao and Gu, 2024), Gated DeltaNet (Yang et al., 2025), and hybrid models such as Jamba (Lieber et al., 2025) offer linear-time alternatives through discretised recurrence and optimised selective memory.

However, a consistent property across these architectures is that channels or hidden units are treated as independent streams. S4 and Mamba explicitly model each channel as an isolated 1D dynamical system with no cross-channel interaction (Gu et al., 2022; Gu and Dao, 2024). Jamba reintroduces attention to compensate for retrieval limitations, but does so at significant cost (Lieber et al., 2025). While LSTMs address vanishing gradients through gating, they do so at the unit level, whereas S4 and Mamba achieve efficiency by abandoning cross-unit coordination entirely, a trade-off that moves further from population-level regulation. This trade-off reveals that architectures achieving computational efficiency do so by eliminating cross-unit coordination that may be necessary for stable population-level representations. Whether this absence contributes to failure modes such as representational collapse is the question this study addresses through population-level coupling via the GF modulation.

2.3 Representational Geometry: Collapse, compression, and the rank debate

DNNs frequently exhibit representational collapse during training, where hidden layer activation occupy an increasingly low-dimensional subspace despite high nominal capacity. Pappayan et al. (2020) document this as "neural collapse" during terminal training phases, while Daneshmand et al. (2020) demonstrate it arises in randomly initialised DNN when neurons cease performing independent computations. Roth and Liebig (2024) extend this finding to graph NNs, suggesting collapse is a broadly shared failure mode rather than an architecture-specific phenomenon.

Whether collapse is inherently detrimental remains debated. Tishby and Zaslavsky (2015) propose the information bottleneck principle, arguing that effective generalisation requires hidden layers to compress input representations by discarding task-irrelevant variance. Under this view, reduced dimensionality is a feature of well-trained networks rather than a pathology. However, Saxe et al. (2019) challenge this empirically, finding no consistent

compression. Hadsell et al. (2006) and Shwartz-Ziv and Tishby (2017) reconcile this by framing dimensionality reduction as a perspective-dependent consequence of learning invariant structure during a fitting-to-compression transition. This suggests that collapse and compression are the same phenomenon viewed from different angles, where success depends entirely on whether discriminative structure is preserved within the reduced subspace.

A similar tension is found in the reservoir computing literature, where ESN's (Jaeger and Haas, 2004) rely on specific dynamical regime that must be carefully tuned to prevent the reservoir's hidden states from collapsing into a singular point or expanding into chaotic divergence. The formalisation that recurrent neural system computation is maximised at the boundary between order and chaos originates with Bertschinger and Natschlager (2004). Recent work suggests that the singular value spectrum of weight matrices tracks this transition directly, distinguishing memorising networks from generalising ones (Yunis et al., 2024). Yet current architectures provide no mechanism to detect or regulate which side of the boundary the network's representations are on during the forward pass. Existing approaches such as elastic weight consolidation (Kirkpatrick et al., 2017) and progressive networks (Rusu et al., 2016) address interference at the weight update level rather than regulating the representational geometry that makes interference likely. This leaves a structural gap: no existing mechanism operates on population-level representational dynamics during the forward pass to regulate whether compression remains discriminative or collapses into redundancy.

The absence of a regulatory mechanism capable of distinguishing beneficial compression from pathological redundancy represents not only a theoretical gap but a measurement gap: without metrics sensitive to the difference between discriminative compression and entropic collapse, neither the presence nor the severity of this failure mode can be detected during training. The dissociation framework introduced in Section 3.3 addresses this directly.

2.4 Temporal stability and the slowness principle

Wiskott and Sejnowski (2002) perspectives on temporal stability comes from Slow Feature Analysis (SFA), which demonstrates that features varying slowly over time tend to capture high-level invariant structure rather than input-level noise. Applied to sequential data, this principle suggests that a signal constrained to evolve slowly will preferentially encode persistent identity-level features rather than rapid low-level transitions. Importantly, SFA assumes that slow features are the most task-relevant; while this holds for classification tasks like digit recognition, it may not generalise to tasks requiring sensitivity to rapid temporal changes.

Continuous-time models such as liquid time-constant networks (Hasani et al., 2021) implement temporal adaptivity through neuron-specific conductance equations, achieving expressive temporal dynamics but at significant computational cost. The leaky integrator formulation adopted here instead applies a single population-level decay, trading unit-level expressiveness for computational efficiency and a shared temporal context.

At the population level, Driscoll et al. (2022) provide a critical empirical anchor for GF modulation through their study of representational drift. By performing longitudinal imaging of thousands of neurons in mice performing a navigation-based memory task, they

demonstrated a fundamental dissociation: while individual neurons shifted their tuning daily (drift), the collective population geometry remained stable. This suggests biological systems maintain a "shared population reference" or a stable manifold that transcends unit-level variability.

This study addresses that gap by formalising this reference as a GF. By anchoring units to this slowly evolving aggregate signal, I test whether the network can achieve the collective stability observed by Driscoll, preventing the "representational collapse" (Papayan et al., 2020) typical of purely local learning. To empirically validate this effect, this study introduces the retention ratio as a diagnostic for temporal invariance. This "reversal memory check" rests on three theoretical pillars: symmetry breaking, where reversing inputs acts as a temporal perturbation to test the GF's resilience; manifold stability, which posits that a stable population geometry should remain invariant regardless of the sequence's directionality; and credit assignment, distinguishing true long-range temporal integration from local sequence tracking. By assessing geometry preservation under inversion, we quantify whether the GF facilitates persistent state spaces independent of learned directional priors.

While existing architectures implicitly generate population activity, none feed this signal back as a modulating force. To isolate this causal effect, a three-condition design is employed: passive, probe (computed but decoupled), and active (full modulation). The contribution is a mechanistic one: that explicit feedback produces a specific dissociable signature, stable ER (Roy and Vetterli, 2007) and elevated temporal autocorrelation (Shi et al., 2023) indicating coordinated binding without drift.

Finally, Interference, measured as the mean correlation between unit activation and the aggregate signal, provides a diagnostic for pathological entrainment. Grounded in the redundancy literature (Daneshmand et al., 2020), this metric tests whether the GF produces the coordination-without-collapse balance predicted by the CTC framework. Together, these metrics form a dissociation framework capable of distinguishing beneficial compression from the pathological redundancy identified in Section 2.3.

3. Methods

This study employs a three-condition comparative design to isolate the causal effect of GF modulation on representational stability. A passive baseline establishes performance without any field computation. A Probe condition computes the field signal but prevents it from influencing the forward pass, establishing what population-level activity looks like without regulatory coupling. An Active condition engages the full modulation mechanism. The Sobol sensitivity analysis is then applied exclusively to the Active condition to characterise which GF parameters most determine representational outcomes. This design allows the study to distinguish between three possible explanations for any observed differences: effects of the architectural structure itself, effects of observing population activity, and effects of active field coupling.

3.1 Network architecture: The global field mechanism

The core computational unit is the “jitteredfeedbackcell”, designed to inject population-level context into individual neuronal updates(the full architectural pipeline is schematized in Figure 2)). At any time step t , the layer's hidden state is represented as $h_t \in \mathbb{R}^n$.

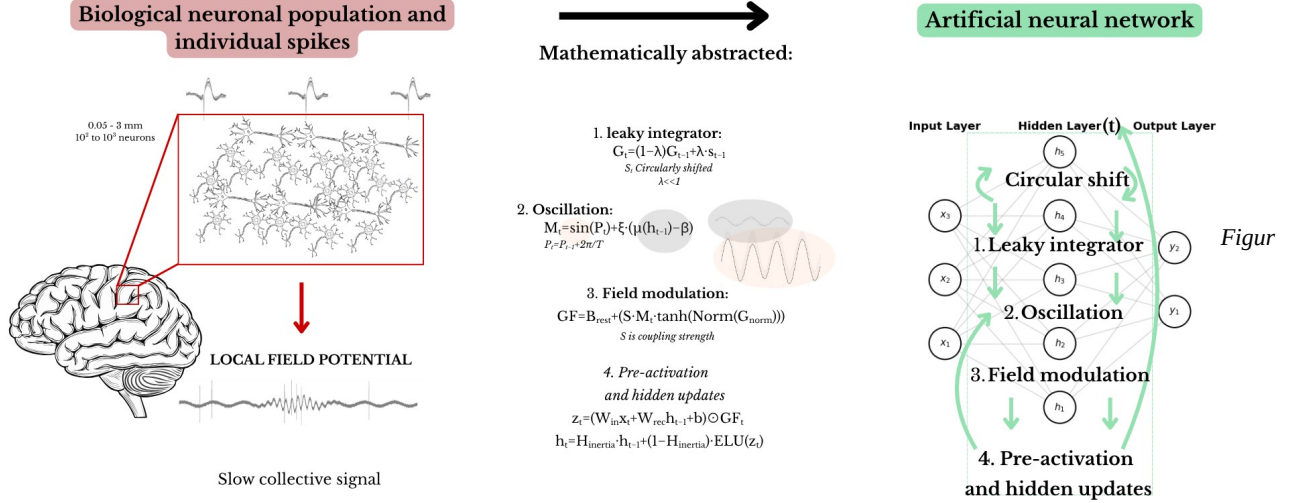


Figure 2, **Architectural schematic of the GF mechanism.** The diagram illustrates the integration of the leaky integrator (slowness), the oscillatory modulation (CTC), and the resulting field-effect on hidden state updates.

3.1.1 The Leaky Integrator and Slowness Principle

The global state G_t is maintained via a leaky integrator applied to a circularly shifted representation of the layer's hidden activity. To prevent trivial self-reinforcement, the source signal s_t is constructed via a circular shift of h_{t-1} followed by reverse-concatenating the upper and lower halves of the hidden vector. This operation serves as a de-correlation mechanism common in signal processing (Oppenheim and Schafer, 2010), ensuring the field integrates a structurally mixed population signal without additional learnable parameters.

$$G_t = (1-\lambda)G_{t-1} + \lambda \cdot s_{t-1}$$

M2: λ represents the decay rate (the temporal low-pass constraint).

The accumulated state is then z-score normalised across the hidden unit dimension independently per sample to stabilise the field signal:

$$G_{\text{norm}} = (G_t - \mu(G_t)) / (\sigma(G_t) + \epsilon)$$

M3: $\mu_t = \text{mean}(G_t), \sigma_t = \text{std}(G_t)$ and $\epsilon > 0$

3.1.2 Oscillatory Modulation and Field Effect

Drawing from the CTC framework, the network generates an internal rhythmic phase P_t governed by a structural period T :

$$P_t = P_{t-1} + 2\pi/T$$

M4: Internal rhythmic phase P_t

A fixed period T is used, as an adaptive oscillator would add parameters and risk co-adapting with learned weights, conflating scaffold and representation. Consistent with CTC, where oscillatory frequencies reflect circuit structure rather than task learning, T is treated as a hyperparameter explored via sobol sensitivity analysis, not a trainable parameter.

The raw oscillator is combined with a biased population signal to create a jittered modulation wave:

$$M_t = \sin(P_t) + \xi \cdot (\mu(h_{t-1}) - \beta)$$

M5: ξ scales the contribution of the mean population activity.

The offset $\beta = 0.1$

To ensure the jitter term exerts a measurable influence on the modulation wave without being dominated the sinusoidal component β is held constant. Additionally β is held constant across all conditions to avoid introducing a free parameter. The resulting field effect GF_t modulates the gain of the pre-activation signal. The resulting field effect GF_t modulates the gain of the recurrent signal. This study uses two distinct modulation regimes:

1. **Global homogeneous modulation:** For image classification benchmarks (sMNIST and psMNIST), the field provides a uniform temporal scaffold derived from the population mean.

$$GF = B_{\text{rest}} + (S \cdot M_t \tanh(\text{Norm}(G_{\text{norm}})))$$

M6: Global homogeneous modulation

2. **Heterogeneous modulation:** For high-interference tasks ($T \sim 1000$ sCIFAR-10 and adding Task), the field is applied via a unit-specific coupling vector γ (referred to as neuron_gain):

$$GF = B_{\text{rest}} + (\gamma_i \cdot S \cdot M_t \cdot \tanh(\text{Norm}(G_{\text{norm}})))$$

M7: Heterogeneous modulation

where B_{rest} is the resting baseline gain and S is the base coupling strength. In the passive condition S is fixed at 0 and in the Probe condition a stop-gradient operation is applied to st before integration, preventing field gradients from influencing weight updates.

3.1.3 Hidden State Update

Unlike standard IndRNN architectures that rely on diagonal weights, this model utilises a dense recurrent weight matrix W_{rec} initialized on an orthogonal manifold (gain=1.0). For the (p)sMNIST-based tasks, the architecture utilizes a three-layer hierarchical stack to facilitate deep feature extraction.

The pre-activation z_t and hidden update h_t are defined as follows:

1. (p)sMNIST (Homogeneous):

$$\begin{aligned} z_t &= (W_{\text{in}}x_t + W_{\text{rec}}h_{t-1} + b) \odot GF_t \\ h_t &= H_{\text{inertia}} \cdot h_{t-1} + (1 - H_{\text{inertia}}) \cdot \text{ELU}(z_t) \end{aligned}$$

M8: Homogeneous pre-activation z_t and hidden update h_t

2. The adding and sCIFAR-10 Task (Heterogeneous):

$$\begin{aligned} z_t &= (W_{\text{in}}x_t + W_{\text{rec}}h_{t-1} \odot GF_t) + b \\ h_t &= H_{\text{inertia}} \cdot h_t + (1 - H_{\text{inertia}}) \cdot \text{LeakyReLU}(z_t) \end{aligned}$$

M9: Heterogeneous pre-activation z_t and hidden update h_t

The unit-specific coupling vector γ_i resembles the fast-weight trace in differentiable plasticity (Miconi et al., 2018), enabling selective pathway modulation. H_{inertia} governs the balance between historical retention and new integration, positioning the network along the order-to-chaos spectrum (Bertschinger and Natschlager, 2004). Activation functions are selected to manage drift: ELU permits negative states in (p)sMNIST, while Leaky ReLU supports high-precision gradients in regression. To prevent runaway excitation, hidden states are clipped to $[-20, 20]$ or, for sCIFAR-10 (Krizhevsky, 2009) and the adding task, a positive-bounded $[0.1, 5]$. This positive range prevents sign ambiguity in the MSE gradient signal, ensuring direction-consistent contributions across long sequences under Leaky ReLU dynamics.

The roles of key GF parameters are summarised in table 1.

Parameter	Conceptual role	Functional impact on hidden states
H_{inertia}	The Stabiliser	Determines "State Persistence." It prevents new inputs from immediately overwriting the network's internal memory.
λ	The Temporal Context	Defines the "History Window." It dictates how long a global event continues to influence future time steps.
S	The Modulation Depth	The "Gating Power." It controls the magnitude of the field's ability to push neurons out of local minima or redundant states.
ξ	The Diversity Driver	The "Stochastic Nudge." Injected jitter prevents pathological synchrony and ensures the network maintains high ER
T_p	The Rhythmicity	The "Pulse." Establishes the frequency for active maintenance, critical for holding information during "silent" delay periods.

Table 1: Key GF parameters and their functional roles in regulating hidden-state dynamics and representational stability.

3.2 Experimental Operational Modes

To isolate the causal impact of the GF on representational geometry, the architecture is evaluated under three regimes (Figure 3):

1. **Passive Baseline:** Standard recurrent architecture with no field computation. The base coupling strength S is fixed at 0, reducing the network to ELU/leaky ReLU-gated recurrent dynamics. H_{inertia} is held constant across conditions to ensure architectural parity.
2. **Probe:** The GF is computed (G_t) and logged, but decoupled from the forward pass.
3. **Active:** The full mechanism is engaged, and the GF actively modulates pre-activation inputs.

This design distinguishes between effects of standard recurrent dynamics, passive population signal observation, and active field coupling.

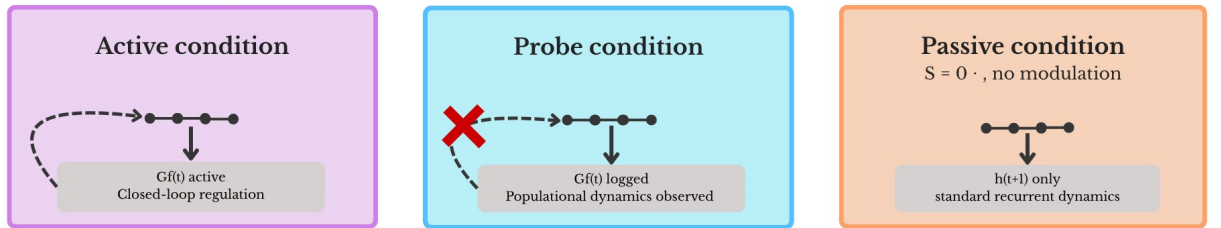


Figure 3, **Experimental operational modes for causal isolation.** The study utilizes a three-condition design to isolate the GF's regulatory effect. Left (Active): Full closed-loop regulation where the GF signal modulates hidden state dynamics. Center (Probe): Open-loop observational mode; the GF signal is computed and logged, but the feedback connection is severed (stop-gradient), allowing tracking of unregulated population drift. Right (Passive): Control baseline utilizing standard recurrent dynamics ($S=0$) without field computation.

3.3 Metric dissociation framework

To quantify the representational diversity and optimisation stability hypotheses outlined in Section 1, hidden state dynamics are captured at the final time step of each epoch, reflecting the settled representational state after full sequence processing. This provides a consistent snapshot across benchmarks. Following the information decomposition approach, these metrics evaluate whether the GF enables functional integration without sacrificing the segregation required for diverse feature representation.

Effective rank: Measures the functional dimensionality (optimal compression) of hidden representations. Following Roy and Vetterli (2007), it is computed as the exponential of the Shannon entropy of the L1-normalised singular value spectrum (σ). By capturing gradual spectral decay beyond hard rank thresholds, ER quantifies representational collapse or bloat:

$$p_i = \sigma_i / \sum_j \sigma_j$$

$$ER = \exp(-\sum p_i \ln p_i)$$

M10: Normalised singular values p_i and entropy-based ER

Synchrony: Calculates the mean off-diagonal absolute Pearson correlation across unit pairs. Following the Communication-through-Coherence framework (Uhlhaas et al., 2009), it captures cross-unit coordination. Pearson correlation is used instead of spectral coherence since hidden states are discrete timestep vectors. It serves as a diagnostic of functional segregation: moderate synchrony reflects coordination, while maximal synchrony indicates loss of unit independence.

Interference: This metric measures the mean absolute Pearson correlation between individual unit activations and the population mean across a batch, computed at the final time step. While Daneshmand et al. (2020) quantify redundancy through rank analysis, unit-to-population correlation serves as a computationally efficient proxy for the same phenomenon, capturing the degree to which units become redundant encoders of the aggregate signal. This aligns with Elhage et al.’s (2022) superposition hypothesis: under strong field coupling, units may be compelled to encode global dynamics at the expense of local feature diversity. High interference values thus signify that units have converged toward redundant copies of the field signal, sacrificing independent computational capacity.

Temporal autocorrelation: Evaluates the persistence of the hidden state across sequential inputs over time (Shi et al., 2023). This acts as a proxy for the network’s smoothing capacity and its ability to maintain a persistent core (Driscoll et al., 2022). High autocorrelation indicates that the GF is successfully acting as a Temporal Anchor, mitigating representational drift and resisting the flicker of high-frequency pixel updates.

Entropy: Measures the distributional spread of hidden unit activations at the final timestep, computed as the Shannon entropy of a histogram approximation to the marginal activation distribution:

$$H = -\sum p(k) \log_2(p(k) + \epsilon)$$

M11: Binned Shannon entropy H of activation distribution measuring representational diversity.

Where $p(k)$ is the normalised count in bin k across 50 fixed bins. High entropy indicates broadly distributed activations (diversity), while low entropy reflects concentration and potential collapse. Following Keller (2019), Shannon entropy provides a parameter-free

summary of distributional complexity without requiring assumptions about the underlying activation distribution.

3.4 Datasets and benchmarking suite

To ensure the GF mechanism is not overfitted to a single dynamical regime, this study evaluates four benchmarks spanning synthetic long-range tasks and sequential image data. Each is selected to focus on a specific aspect of representational stability rather than maximise performance.

SMNIST are MNIST (Yann LeCun et al., 1998) digits flattened into 784-step sequences. The network must classify only at the final time step, testing long-range credit assignment and hidden state stability. sMNIST is retained as a canonical benchmark for direct comparison with prior recurrent work (Le et al., 2015; Hochreiter and Schmidhuber, 1997).

PsMNIST are MNIST images presented as sequences with a fixed random pixel permutation applied across all samples. This destroys local spatial correlations, forcing reliance on global temporal integration and testing gradient stability under degraded input order. PsMNIST acts as a controlled perturbation of sMNIST, isolating dependence on short-range structure; performance differences directly reflect reliance on local patterns versus global temporal dynamics.

The adding task is a two length- $(L=1000)$ sequences are input: one with random values in $([0,1])$, the other a binary marker sequence with two positions (early and mid-sequence). The model must output the sum of the marked values, requiring selective retention across long delays without interference or decay. Unlike copy tasks, this demands functional integration and interference resistance, serving as a direct test of gradient stability and independent unit dynamics under sustained input pressure. This serves as a direct stress test of the GF’s hypothesized ability to maintain independent unit computations. By holding the sequence length at $L=1000$, the task demands gradient stability comparable to SOTA architectures (see Table 2).

The sCIFAR-10 dataset is derived from CIFAR-10 (Krizhevsky, 2009) by serialising each 32×32 RGB image into a 1,024-step sequence, where each timestep corresponds to a single 3-channel pixel. Increased dimensionality and nonlinearity make this the primary stress test for ER and resistance to representational collapse.

An initial plan to evaluate real-world intracranial EEG signals was de-scoped to ensure computational feasibility and maintain a strict focus on core-controlled benchmarks

Model	Benchmark	Parameters	T	Academic Source
IndRNN	Adding task	~17 000	1 000	Li et al. (2018)
GF (h64) indRNN	Adding task	4 417	1 000	
GF (h128) indRNN	Adding task	17 025	1000	
AntisymmetricRNN (gated)	sCIFAR-10	~37 000	1024	Chang et al. (2019)
AntisymmetricRNN	sCIFAR-10	~36 000	1024	Chang et al. (2019)
GF (h64) indRNN	sCIFAR-10	21 706	1024	
LTC	sMNIST	4 682	784	Hasani et al. (2021)
GF (h16)	(p)sMNIST	1 514	784	
GF (h32)	(p)sMNIST	5 578	784	

Table 2: Low (SOTA) parameter count model comparison on all the benchmarks, reporting parameters, sequence length and academic source.

3.5 Global Sensitivity Analysis

The Sobol sensitivity analysis is applied exclusively to the active condition, with passive and probe serving as fixed reference anchors. The goal is not condition comparison, but characterisation of the internal stability landscape of the active regime, identifying which GF parameters govern whether modulation improves or degrades representational geometry relative to these baselines. The analysis therefore asks which parameters control the coordination–collapse trade-off hypothesised for the GF.

Table 3 details the hyperparameter boundaries applied across these benchmarks. Parameters such as the oscillatory period T and decay rate λ are scaled relative to sequence length to ensure the GF operates in an appropriate dynamical regime for each task.

Sensitivity indices are computed jointly over five outputs: classification accuracy, ER, synchrony, entropy, temporal autocorrelation, and interference. This reveals whether parameters influence performance and geometry independently or jointly. First-order indices (S1) quantify isolated effects, total-order indices (ST) capture full interaction-inclusive influence and are the primary measure reported, and second-order indices (S2) capture pairwise parameter interactions.

Benchmark parameter	sMNIST	psMNIST	Adding task	sCIFAR-10
Fixed variables				
Epochs	10	10	150	10
Architecture	RNN	RNN	IndRNN	indRNN
Layers	3	3	1	3
Hidden	32	16 & 32	128	64
Sweeps	784	784	96	384
Data %	10	10	100	10
T	784	784	1000	1024
Batch size	512	512	256	128
Activation	Elu	Elu	Leaky Relu	Leaky Relu
B_{rest}	1	1	1	1
	-20	-20	0.1	0.1
Clipping	20	20	5	5
Learning rate (a)	1e-3	1e-3	2e-4	5e-3
Minimum a			1e-6	1e-6
Patience			10	2
Monitor			Val loss	Val loss
Boundary parameters				
	0.01	0.01	0.0001	0.001
λ	0.05	0.05	0.0015	0.05
	0.01	0.01	0.02	0.01
S	0.99	0.99	0.15	0.99
	0.1	0.1	0.01	0.1
ξ	1.2	1.2	0.10	1,2
	T/12	T/12	T	T/13
T_p	T	T	T·5	t/2
	0.01-	0.75	0.92	0.01
$H_{inertia}$	0.99	0.99	0.99	0.99

Table 3, The sobol sweep mapped the internal mechanics of the GF across five architectural dimensions ($D=5$): lambda slow (λ), base coupling strength (S), jitter scale (ξ), oscillatory period (T_p), and temporal inertia ($H_{inertia}$), across several datasets namely (p)sMNIST, the adding problem and sCIFAR-10. The bounds can be read from the table above. The Saltelli formula gives $N(2D+2) = 12N$ evaluations. At $N=64$ this yields a maximum of 768 independent training sessions.

3.6 Training Procedure

The training loop is designed to maintain architectural parity across all three conditions, ensuring that differences in representational metrics are attributable to the GF rather than parameter count or optimisation dynamics.

For classification tasks ((p)sMNIST and sCIFAR-10), the loss is Sparse Categorical Cross-Entropy. For the adding task, Mean Squared Error (MSE) is used. All conditions are trained with Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$). Adam is preferred over SGD with momentum due to its adaptive per-parameter learning rates, which better handle the non-stationary gradient scales arising in full BPTT, particularly across long sequences and between recurrent and field parameters. The learning rate is fixed to avoid confounding sensitivity analyses.

Full Backpropagation Through Time (BPTT) is applied over the entire sequence length without truncation (784 for sMNIST/psMNIST, 1000 for adding, 1024 for sCIFAR-10). This ensures the temporal anchoring mechanism is evaluated under complete gradient flow. Gradient norm clipping (Pascanu et al., 2013) with threshold 1.0 is applied to prevent exploding gradients while preserving signal propagation. Table 4 summarises the fixed training specifications across all benchmarks and conditions.

Component	Specification / Value	Note
Framework	TensorFlow 2.1x (Keras API)	Custom JitteredFeedbackCell implementation
Optimizer	Adam ($\beta_1=0.9, \beta_2=0.999$)	Standard for non-stationary RNN dynamics
Loss Function	Sparse Categorical Cross-Entropy	MSE used for Adding Task regression
BPTT Depth	Full (784 to 1024 steps)	No truncation to test long-range credit assignment
Gradient Clipping	Global Norm Threshold = 1.0	Prevents exploding gradients in deep recurrence
Hardware	NVIDIA GPU (CUDA/cuDNN)	Mixed precision (float32) for precision
Reproducibility	Global Seed = 42	Reset via reset_seeds() before every run

Table 4: specification summary

3.7 Computational Resources and Feasibility

The methodology intentionally avoids over-parameterization, allowing local execution on a consumer workstation (Intel i5, 8GB RAM, RTX 3050). Storage requirements are minimal, as the selected datasets are highly structured and model checkpoints for these low-parameter networks require negligible disk space.

Due to the memory demands of Backpropagation Through Time (BPTT) over long sequences ($L=784$), model states and GPU memory are aggressively cleared between each of the generated Sobol configurations to prevent Out-Of-Memory failures.

3.8 Architectural Control Across Conditions

The three experimental conditions are kept parameter-identical to isolate the causal effect of the GF. In the passive condition, the coupling strength S is fixed at 0, reducing the model to standard ELU-gated recurrent dynamics for sMNIST and psMNIST, and Leaky ReLU-gated dynamics for the adding task and sCIFAR-10, with no field computation. In the probe condition, the GF is computed and logged, but a stop-gradient is applied so it does not influence the forward pass. In the active condition, the full closed-loop mechanism is engaged, with GF modulating hidden-state gain in real time.

All conditions use identical initialisation: orthogonal recurrent weights (Saxe et al., 2013) and Glorot uniform input weights (Glorot & Bengio, 2010). Orthogonal initialisation preserves gradient norms via unit singular values, reducing early instability before GF effects emerge, while Glorot scaling maintains activation variance at the input layer.

3.9 Reproducibility

The implementation uses TensorFlow 2.19.1 with custom Keras layers for the JitteredFeedbackCell. Experiments are executed on an NVIDIA GPU with XLA just-in-time compilation selectively disabled where numerical instability was observed. All stochastic elements, including weight initialisation, data shuffling, and any dropout operations, are fixed to a global seed of 42 via a `reset_seeds()` utility applied before each model initialisation, ensuring exact reproducibility across configurations.

3.10 Evaluation Protocol

Representational metrics (ER, synchrony, interference, and temporal autocorrelation) are evaluated at the final time step of the final batch of every epoch, capturing the settled state of the hidden manifold after each full pass through the training data. Classification accuracy and regression error are computed on a dedicated validation split of 10% of the training data after every epoch. Alongside a final test split of 10% at the end of the training.

3.11 Statistical analysis and sobol convergence

To address the known convergence problem at low sample counts in Sobol analysis, the Saltelli sampler is used throughout, which generates $N(2D+2)$ model evaluations per sweep rather than N alone. For the 5D sweep at $N=8-64$, this produces 96-768 evaluations. Stability of the sensitivity indices is assessed by comparing results across sampling densities within each sweep; convergence is indicated when the ranking of parameter importance is stable across N values. Variance in the sensitivity indices across configurations is reported alongside the indices themselves to indicate reliability.

3.12 Ethics and data licensing

This study does not involve the collection of new human or animal data. All datasets used are pre-existing, publicly available resources accessed strictly in accordance with their respective licences and terms of use.

The standard machine learning benchmarks, MNIST (LeCun et al., 1998), CIFAR-10 (Krizhevsky and Hinton, 2009), and the synthetic adding task, carry no ethical concerns beyond standard academic use. These datasets contain no personally identifiable information and were designed explicitly for open research purposes.

No human participants were recruited, no deception was employed, and no personally identifiable data was handled at any stage of this research. The study poses no foreseeable risk of harm to individuals or groups. The primary ethical obligation is adherence to the licensing terms of the datasets used, which has been observed throughout.

3.13 Limitations

Several methodological constraints should be considered when interpreting the results.

All experiments use a single random seed (42), ensuring reproducibility but preventing estimation of initialisation variance. The shared-seed, parameter-identical design across conditions partially insulates within-condition comparisons from initialisation artefacts, meaning observed differences between passive, probe, and active reflect architectural effects rather than weight initialisation. However, absolute performance estimates remain unvalidated across initialisations and should be confirmed via multi-seed replication. As a partial mitigation, bootstrap confidence intervals on the Sobol total-order indices could have been computed across resampled evaluation subsets without requiring full multi-seed replication, providing a bound on index reliability within the existing runs

Experiments are run on consumer hardware (Intel i5, 24GB RAM, RTX 3050 8GB VRAM), limiting hidden size, batch size, epoch count and Sobol coverage. As a result, GF behaviour is characterised only in low-parameter regimes and may not generalise to larger models.

Sobol sensitivity analysis is applied only to the active condition. Sensitivity may differ across tasks, so cross-benchmark generalisation is not established.

Failed runs are retained in Sobol analysis: assigned 999.0 in the Adding Task and 0.0 in sCIFAR-10 and (p)sMNIST. This ensures instability regions contribute to sensitivity estimates rather than being excluded.

Finally, full BPTT is used without truncation, improving temporal fidelity but limiting computational throughput, which reduces coverage of hyperparameter space at longer sequence lengths.

4. Results

This section presents the empirical findings of the three-condition comparative design, followed by a global sensitivity analysis characterising how the GF parameters influence representational outcomes.

To enable reproduceable Sobol sampling, all benchmarks are conducted under single-seed conditions. Results are therefore interpreted as architectural trends in representational dynamics rather than population-level performance estimates

4.1 Three-Condition Comparison

The most significant performance gain observed in this study occurred in the high-complexity sCIFAR-10 benchmark. The Active condition achieved a test accuracy of 46.86%, representing a 4.6-fold improvement over the passive baseline (10.00%), which failed to exceed chance levels (Table 5). While absolute accuracy remains below SOTA baseline, the GF is notably more parameter-efficient, especially on psMNIST and sCIFAR-10 where it outperforms SOTA models in its weight class in accuracy per weight (Table 6)

A more nuanced trend emerged in the MNIST benchmarks:

- sMNIST (Sequential): The Active condition outperformed the Passive baseline by 4.08 percentage points (80.16% vs. 76.08%).
- psMNIST (Permuted): Performance converged across all conditions (approximately 86%).

Benchmark	Epochs	Hidden	Size	Passive	Probe	Active
sMNIST(%)	25	32	5 578	76.08	75.06	80.16
psMNIST (%)	50	32	5 578	86.62	86.62	86.08
psMNIST (%)	50	16	1 514	54.60	56.61	55.07
Adding (mse)	150	128	17 025	.0178	.0166	.0271
Adding (mse)	250	64	4 417	.0247	.0258	.0375
sCIFAR-10 (%)	30	64	21 706	10.00	36.10	46.86

Table 5: Test accuracy results comparison across benchmarks. With random seed = 42.

Model	Hidden	Size	Benchmark	Accuracy	Weight efficiency
Antisymmetric RNN (gated)	256	37 000	sCIFAR-10	62.2%	1.68e-3

GF	32	21 706	sCIFAR-10	46.9%	2.16e-3
LTC	32	4 682	sMNIST	98.4%	21.02e-3
GF	32	5 578	sMNIST	80.2%	14.37e-3
Antisymmetric RNN	128	10 000	psMNIST	95.8%	9.58e-3
GF	32	5 578	psMNIST	86.1%	15.43e-3

Table 6: Evaluation accuracy and weight efficiency (accuracy divided by the models size) on sCIFAR-10 and (p)sMNIST

4.1.1 Temporal Anchoring and Reversal Analysis

To evaluate the rigidity of the learned representations, I utilised retention ratio (= reversal test score% / Test accuracy %) and $MSE\Delta\%$ for the Adding Task.

- sCIFAR-10: The active condition showed the most severe anchoring, with a ratio of 0.54 compared to the passive ratio of 1.00 (Table 7).
- Adding Task: In contrast, the adding task exhibited near-zero $\Delta\%$ changes across all conditions.

Benchmark	Hidden	Passive	Probe	Active
Adding (mse $\Delta\%$)	128	.0165 5.4	.0173 2.4	.0255 0.2
Adding (mse $\Delta\%$)	64	.0244 -1.5	.0236 -0.4	.0366 0.4
sCIFAR-10 (% ratio)	64	10.00 1.00	24.48 0.68	25.49 0.54

Table 7: Reversal scores and test to reversal accuracy ratio
Ratio = reversal test score% / Test accuracy % on 1000 samples
 $\Delta\% = ((MSE_{reversed} - MSE_{test}) / MSE_{test}) * 100$ on 5000 samples
Both with random seed = 43 to prevent data leakage from the training set.

4.1.2 Visualising internal dynamics

Figures 4 and 5 show hidden state heatmaps for sCIFAR-10. The passive mode exhibits high-density activation washout, with values saturated near the leaky-ReLU ceiling. The active mode shows sparser, more distinct activation bursts. In the adding task (Figures 6-7), the active mode displays roughly double the activation magnitude compared to passive and probe conditions, while maintaining a similar temporal pattern: activity plateaus after the first input and rises after the second, independent of sequence reversal.

Figure 8 shows that high-strength ($S \approx 0.81$) GF modulation in psMNIST ($h=32$) produces elevated synchrony and a clear rhythmic waveform.

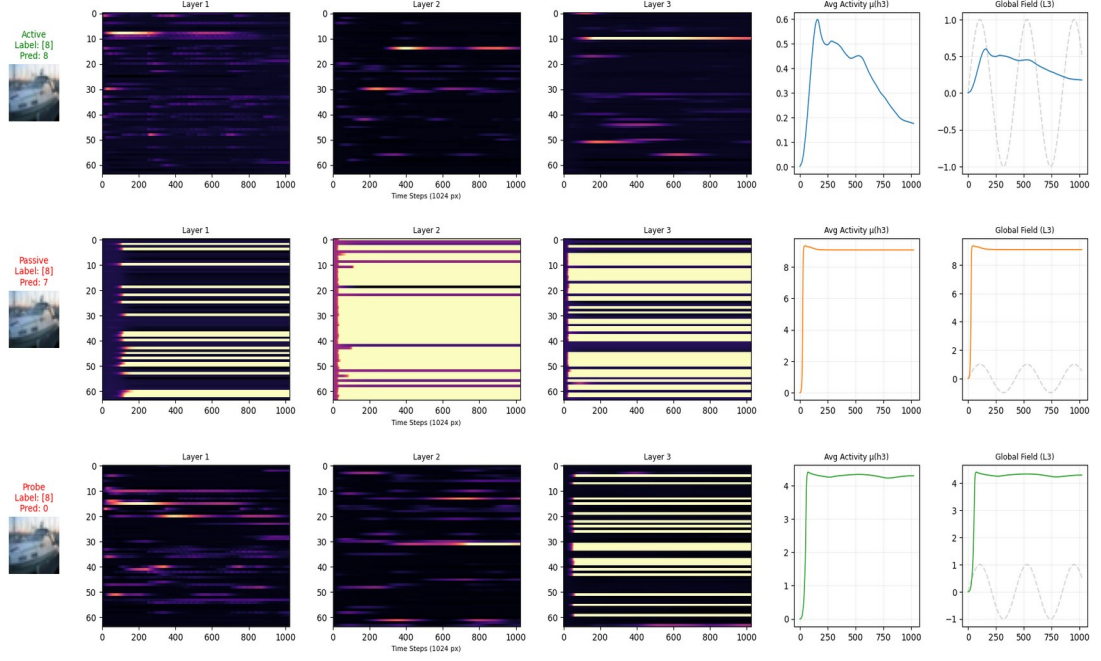


Figure 4, **Hierarchical hidden state dynamics** for sCIFAR-10. Visual representation of the three-layer stacked RNN across modulation conditions. (Left) The original 32 times 32 CIFAR-10 input image. (Center) hidden state heatmaps for Layers 1 through 3, where the y-axis represents individual neurons ($H=128$) and the x-axis represents the 1024 temporal steps. (Right) The population mean activity $\mu(h_3)$ of the terminal layer and its reconstructed interaction with the oscillatory GF.

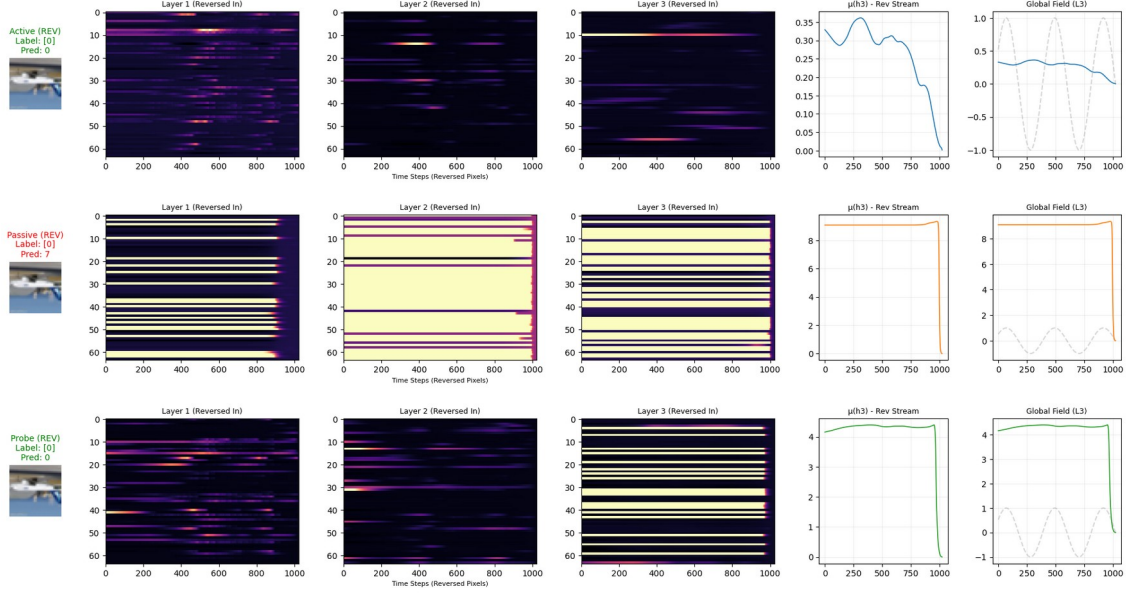


Figure 5 Hierarchical hidden state dynamics for reversed input of sCIFAR-10. Visual representation of the three-layer stacked RNN across modulation conditions. (Left) The reversed 32 times 32 CIFAR-10 input image. (Center) hidden state heatmaps for Layers 1 through 3, where the y-axis represents individual neurons ($H=128$) and the x-axis represents the 1024 temporal steps. (Right) The population mean activity $\mu(h_3)$ of the terminal layer and its reconstructed interaction with the oscillatory GF.

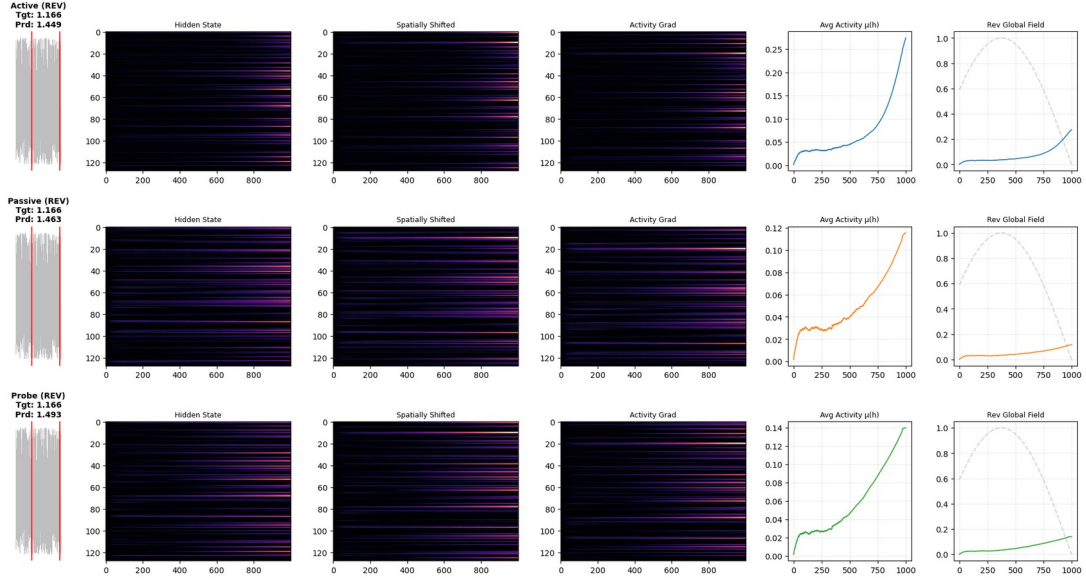


Figure 6, *Neural Dynamics and Global Field Modulation in the reversed adding task ($T=1000$)*. Comparative visualisation of sequence processing across Passive, Probe, and Active modulation conditions. (Left) Dual-channel input representation displaying the uniform value stream (gray) and the discrete binary mask triggers (red vertical lines) that define the indices to be summed. (Center) Hidden state heatmaps illustrating the internal representational evolution: the raw hidden state (h), the spatially shifted source signal used for feedback, and the resulting activity gradient. The y-axis represents the neuron index ($H=128$), while the x-axis tracks the 1000-step temporal duration. (Right) The population mean activity $\mu(h)$ and its reconstructed relationship with the oscillatory GF.

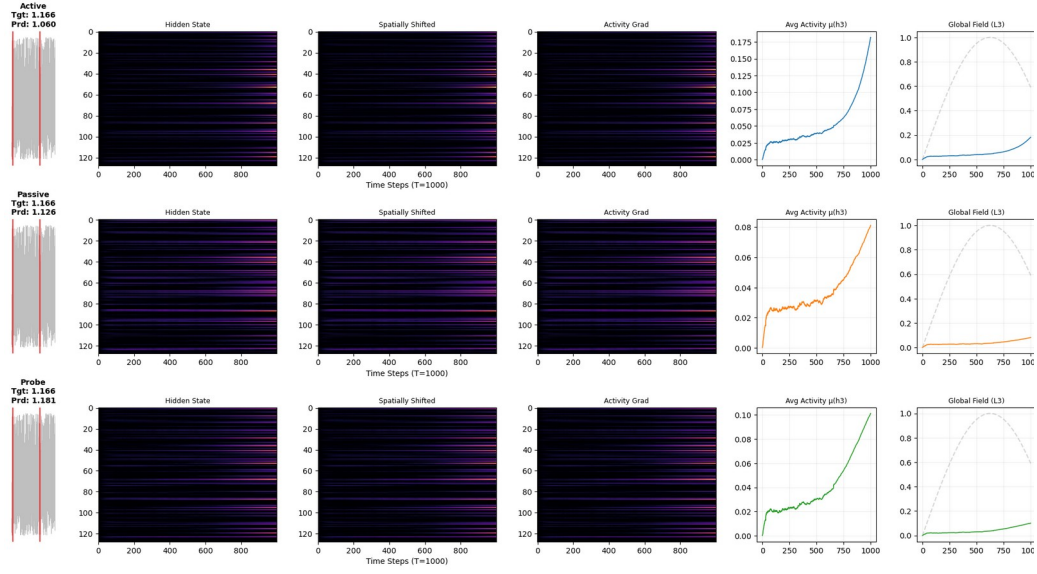
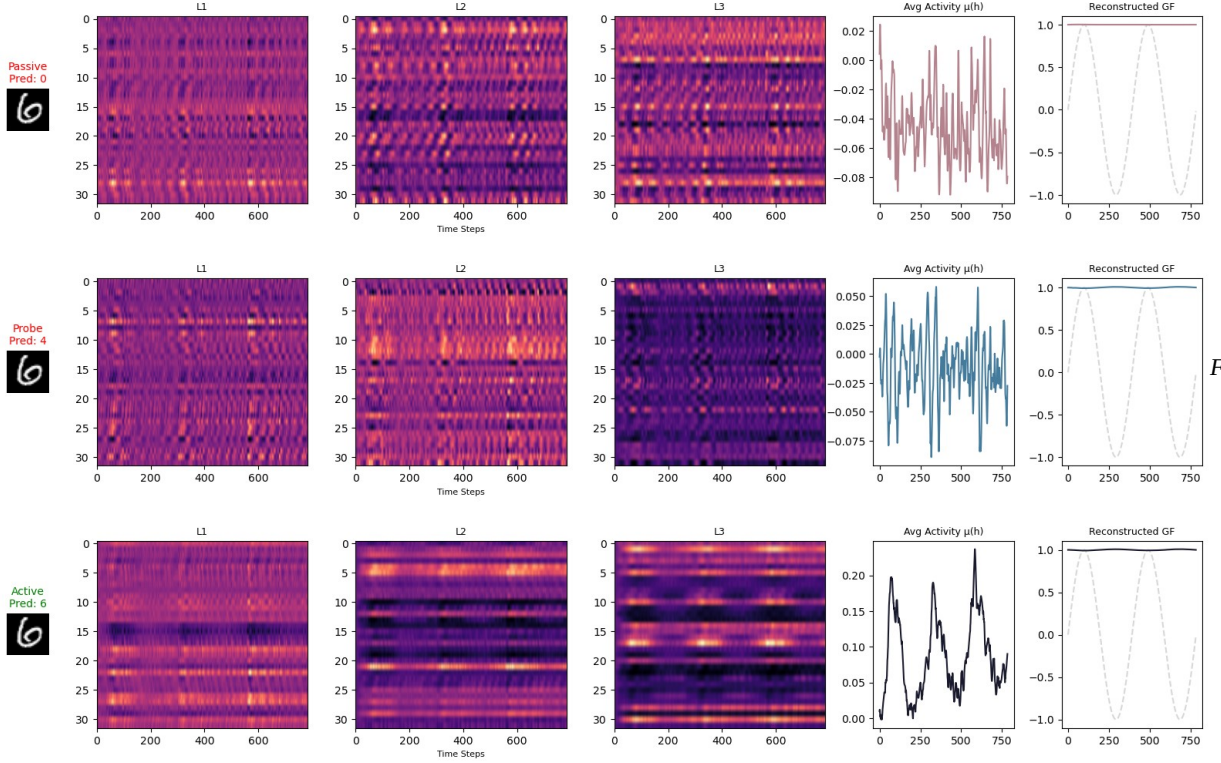


Figure 7, **Neural Dynamics and Global Field Modulation in the Adding Task ($T=1000$)**. Comparative visualization of sequence processing across Passive, Probe, and Active modulation conditions. (Left) Dual-channel input representation displaying the uniform value stream (gray) and the discrete binary mask triggers (red vertical lines) that define the indices to be summed. (Center) Hidden state heatmaps illustrating the internal representational evolution: the raw hidden state (h), the spatially shifted source signal used for feedback, and the resulting activity gradient. The y-axis represents the neuron index ($H=128$), while the x-axis tracks the 1000-step temporal duration. (Right) The population mean activity $\mu(h)$ and its reconstructed relationship with the oscillatory GF.



Figure

8: Neural Dynamics in the psMNIST Benchmark. Comparative visualization of a 3-layer stacked RNN architecture ($H=32$) processing a permuted 784-pixel sequence. **(Left)** The original 28x28 input digit (e.g., Digit 6) before permutation. **(Center)** Hidden state heatmaps for Layers 1, 2, and 3, displaying the evolution of activity across 784 temporal steps. The randomized pixel order is reflected in the high-frequency transitions within the early layers. **(Right)** The population mean activity $\mu(h)$ and the reconstructed GF for the terminal layer.

4.1.3 Metric analysis

In the psMNIST benchmark, hidden state dynamics drift toward lower synchrony and higher interference, while the sequentialised trajectories exhibit a distinct reduction in interference (Figure 10). Across most benchmarks, models move synchronously toward the "high rank specialisation" frontier; however, sCIFAR-10 presents a notable exception where only the active condition successfully maintains this progression. Both active and passive models utilise available ER during task convergence (Figure 11).

Under homogeneous fields, trajectories remain in a high-entropy regime while moving toward lower loss (Figure 9). In sMNIST, the active condition maintains a higher-entropy trajectory than both passive and probe throughout training (Figure 9).

In the Adding Task, a divergence in representational strategy emerges. The active condition follows a downward-right trajectory maintained within a positive entropy regime (Figure 9). Conversely, passive and probe modes transition into negative entropy realms. This disparity is statistically supported by the SOBOL sensitivity analysis, which reveals that the heavy sensitivity S has on entropy ($S_1 = -3.21$, $S_t = 15.63$) (Table 11). The coordination portraits exhibit "stochastic wandering," where hidden units remain highly plastic rather than settling into a fixed rhythmic scaffold.

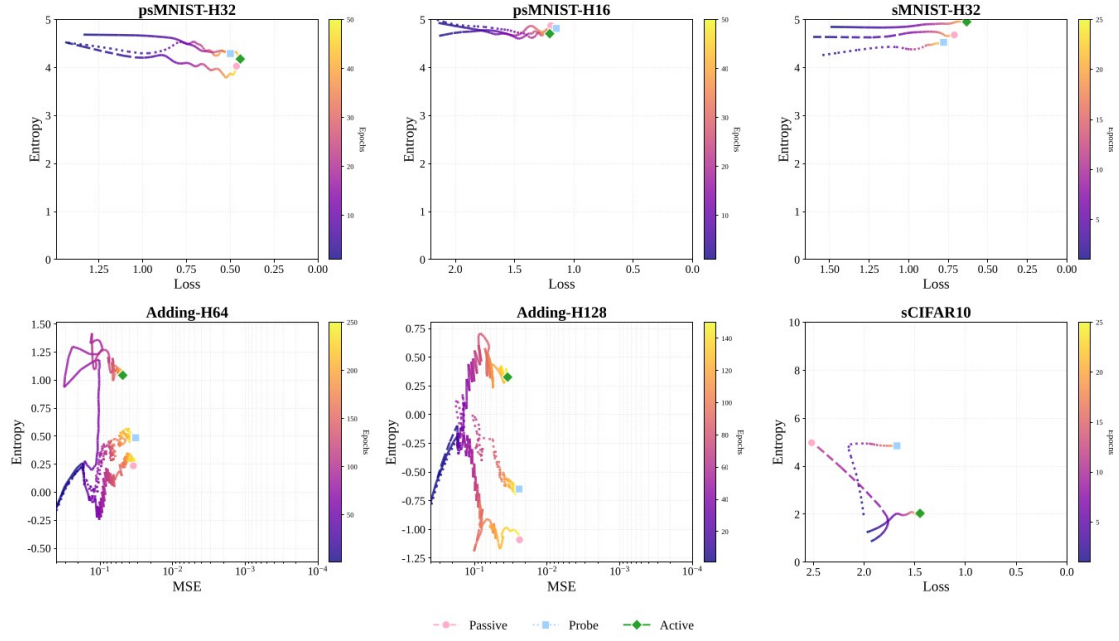


Figure 9: **Phase Portraits of Hidden State Entropy vs. Loss.** This figure maps the entropy activations against the optimisation objective (Cross-Entropy Loss or MSE) over training epochs.

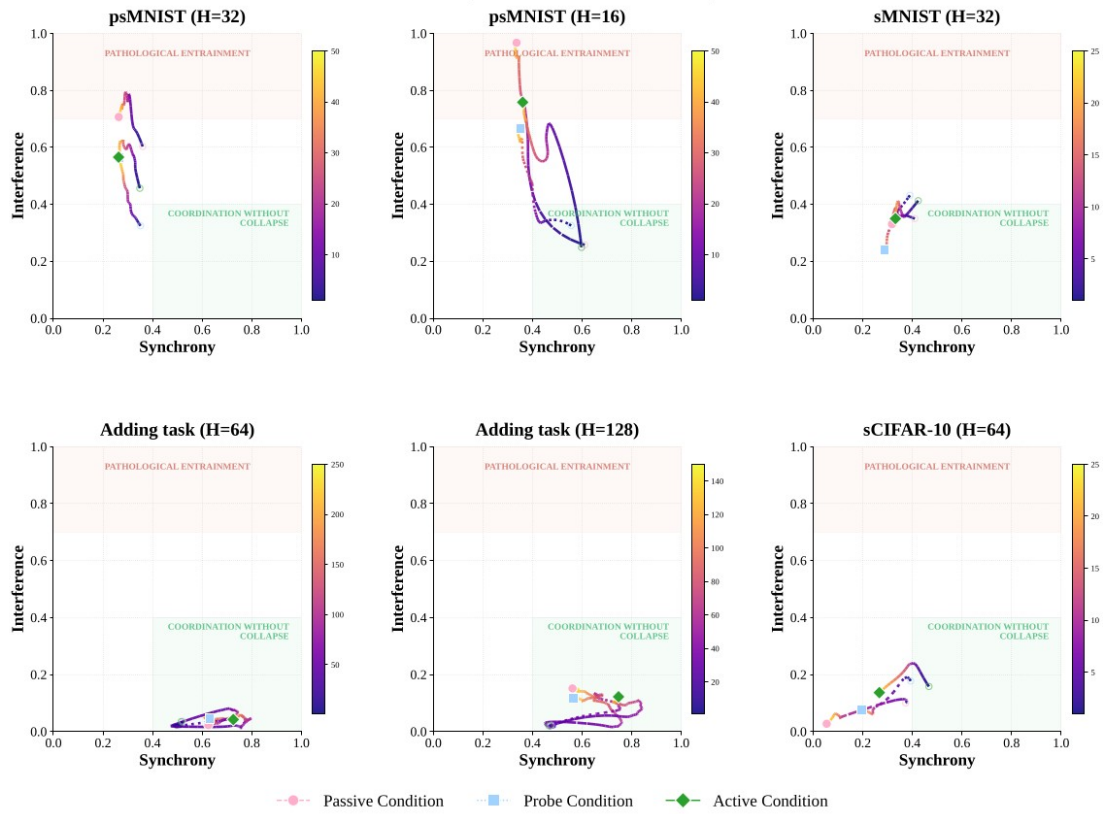


Figure 10: **Epoch Trajectories through Synchrony-Interference Space.** This grid plots the synchrony and interference. The red zone denotes pathological entrainment, while the green zone denotes coordination without collapse.

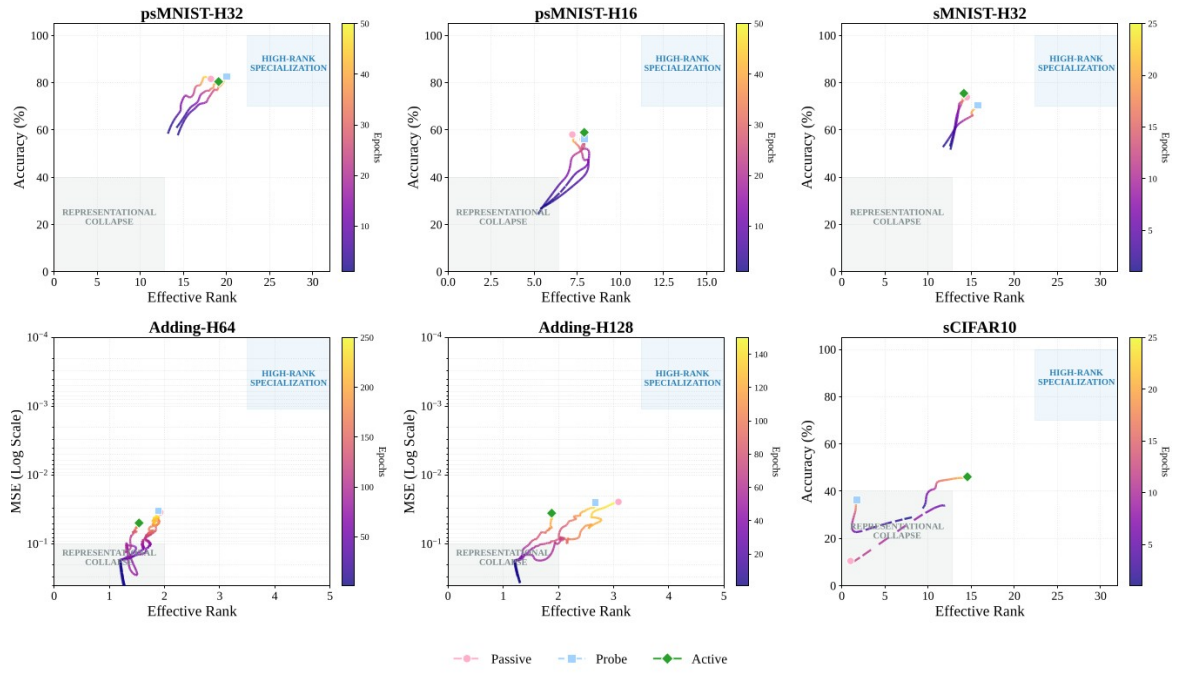


Figure 11: **ER vs. performance trajectories.** These plots track the exponential of the ER against accuracy or MSE. The Representational Collapse region (grey) denotes a functional dimensionality below the task-specific threshold required for feature segregation.

4.2 Sobol Sensitivity Analysis

A Sobol sensitivity analysis was conducted, with total-order (ST) and first-order (S1) indices reported in Tables 7 through 11. In the H32 psMNIST task (Table 8), the interaction between field decay λ and ξ yields a dominant total sensitivity (ST = 1.10) for ER, despite negligible first-order effects. H_{inertia} exhibits high ST values for accuracy in sMNIST (0.90, Table 10) and sCIFAR-10 (1.02, Table 12), while in the adding task (Table 11), field strength drives entropy with a strong total sensitivity (ST = 15.63). Second-order interaction matrices (Table A4) show strong couplings between S and λ , and between H_{inertia} and T_p , while Table A3 highlights an interaction between H_{inertia} and ξ (1.687) influencing interference. In psMNIST, synchrony increases beyond ~ 0.92 alongside declines in accuracy and rank (Figs. 12–14), whereas the adding task shows a stable regime between 0.94–0.97 with high rank, low synchrony, and low MSE (Figure 16). In sMNIST, H_{inertia} displays high total sensitivity across most metrics (Figures. 15, 7), while in sCIFAR-10, values above ~ 0.96 correspond to reduced synchrony alongside peak accuracy and increased rank (Figure 17). The Active condition outperforms the Passive baseline in sCIFAR-10 and sMNIST, while performance converges across conditions in psMNIST.

Metric	λ (S_1 S_T)	H_{inertia} (S_1 S_T)	S (S_1 S_T)	T_p (S_1 S_T)	ξ (S_1 S_T)
Accuracy	-0.16 0.47	-0.068 0.65	-0.12 0.60	-0.16 0.89	-0.15 0.11
Rank	0.036 0.21	0.73 0.89	-0.23 0.22	-0.45 0.47	0.21 0.34
Interference	0.11 0.13	0.097 0.69	-0.41 0.37	-0.76 0.62	-0.21 0.21
Entropy	-0.18 0.28	0.055 0.53	0.43 0.41	0.19 0.38	-0.076 2.00
Synchrony	-0.089 0.40	-0.29 0.71	-0.22 0.47	-0.44 0.79	-0.28 0.21

Table 8: H16 psMNIST

Metric	λ (S_1 S_T)	H_{inertia} (S_1 S_T)	S (S_1 S_T)	T_p (S_1 S_T)	ξ (S_1 S_T)
Accuracy	0.063 0.57	0.035 0.72	0.35 0.62	-0.073 0.61	-0.20 0.76
Rank	-0.037 1.10	0.11 0.67	-0.028 0.94	-0.069 0.32	0.12 1.10
Interference	-0.027 0.61	0.21 0.61	0.47 0.68	0.16 0.33	0.05 0.84
Entropy	0.075 0.50	0.10 0.77	0.04 0.59	0.22 0.53	0.25 0.77
Synchrony	0.14 0.66	0.016 0.73	0.21 0.84	0.076 0.71	0.011 0.82

Table 9: H32 psMNIST

Metric	$\lambda (S_I S_T)$	$H_{inertia}(S_I S_T)$	$S(S_I S_T)$	$T_p(S_I S_T)$	$\xi(S_I S_T)$
Accuracy	0.04 0.26	0.72 0.90	0.14 0.29	-0.02 0.22	0.16 0.39
Rank	-0.10 0.42	0.54 0.66	0.13 0.38	0.08 0.36	-0.10 0.55
Interference	0.04 0.39	-0.05 0.54	0.07 0.44	0.27 0.84	0.15 0.48
Entropy	-0.06 0.19	-0.01 0.67	0.12 0.45	-0.03 0.27	0.09 0.65
Synchrony	-0.14 0.30	-0.35 0.76	0.11 0.47	-0.15 0.26	-0.04 0.71

Table 10: H32 sMNIST

Metric	$\lambda (S_I S_T)$	$H_{inertia}(S_I S_T)$	$S(S_I S_T)$	$T_p(S_I S_T)$	$\xi(S_I S_T)$
MSE Loss	0.26 0.79	1.07 1.58	0.32 0.36	0.27 0.30	0.22 0.28
Rank	0.60 0.34	0.97 0.72	-0.34 0.85	0.74 0.39	0.23 0.40
Interference	-0.05 0.27	0.61 1.20	-0.24 1.50	-0.10 0.73	-0.20 0.24
Entropy	1.00 0.82	0.70 0.82	-3.21 15.63	1.17 0.89	0.20 0.90
Synchrony	-0.39 0.27	0.86 0.66	1.73 13.36	0.38 0.70	0.22 0.19

Table 11: H128 Adding

Metric	$\lambda (S_I S_T)$	$H_{inertia}(S_I S_T)$	$S(S_I S_T)$	$T_p(S_I S_T)$	$\xi(S_I S_T)$
Accuracy	-0.06 0.59	0.70 1.02	0.01 0.33	-0.09 0.56	0.11 0.47
Eff. Rank	-0.09 0.64	0.15 0.75	-0.03 0.45	-0.03 0.77	-0.10 0.57
Interference	-0.22 0.29	-0.12 0.51	-0.25 0.61	-0.14 0.48	-0.19 0.44
Entropy	-0.54 1.36	-0.29 0.98	-0.72 0.77	-0.81 1.73	-0.77 1.88
Synchrony	-0.00 0.47	0.11 0.73	-0.08 0.48	0.01 0.69	0.08 0.67

Table 12: H64 sCIFAR-10

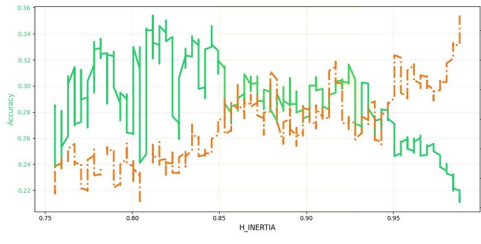


Figure 12: psMNIST h32

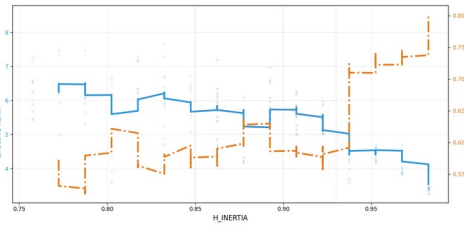


Figure 13: psMNIST h16

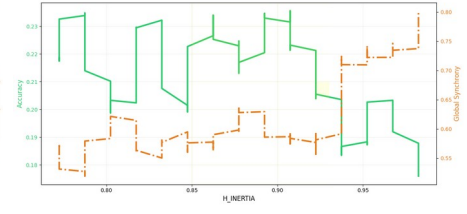


Figure 14: psMNIST h16

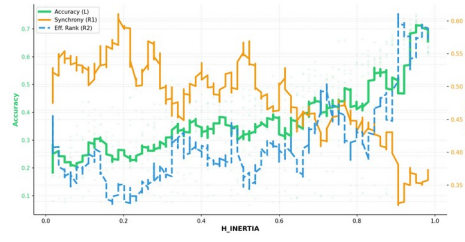


Figure 15: sMNIST h32

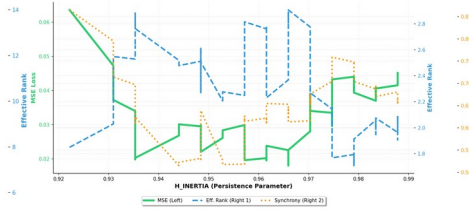


Figure 16: Adding task h128

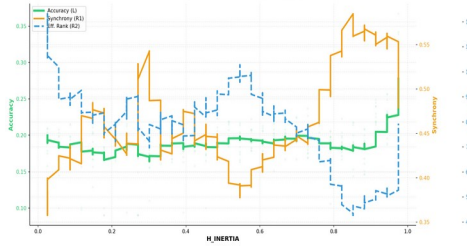


Figure 17: sCIFAR-10

Chapter 5: Discussion

5.1 Overview

The results show that the GF mechanism does not act as a uniform performance enhancer, but as a dynamic, task-dependent regulatory layer whose effects are mediated by parameter interactions rather than any single hyperparameter. Three findings structure this discussion. First, the GF produces a task-conditional performance profile: a clear accuracy benefit in sequential classification, while converging to similar performance on psMNIST and underperforming on the adding task. Second, H_{inertia} emerges as the dominant parameter across benchmarks, with Sobol analysis identifying a bifurcation threshold beyond which network dynamics undergo a qualitative phase transition. Third, representational geometry results, evaluated through the five-metric framework, partially support the coordination-without-collapse hypothesis while revealing boundary conditions on when population-level feedback is beneficial.

5.2 The H_{inertia} bifurcation and the edge-of-chaos parallel

The most consistent finding across all benchmarks is the existence of a critical H_{inertia} threshold beyond which network dynamics undergo a fundamental phase transition (Figs. 12-17). This is consistent with the network transitioning from over-coupled entrainment into the productive metastable regime. This bifurcation structure is consistent with the edge-of-chaos hypothesis in recurrent neural systems, which proposes that maximal computational capacity is achieved at the boundary between ordered (high synchrony, low variance) and chaotic (low synchrony, high variance) dynamical regimes (Bertschinger and Natschlager, 2004). What the GF contributes is not simply proximity to this boundary but an active regulatory mechanism that modulates where along this spectrum the network operates. The high Sobol total-order indices for H_{inertia} on accuracy (Table 8-12), with ST exceeding S1 substantially in all cases, confirming that these effects arise through interaction with other GF parameters rather than H_{inertia} acting in isolation. This places the GF closer to Bertschinger and Natschlager's (2004) characterisation of computational phase transitions as emergent from parameter interactions than to a simple threshold mechanism.

5.3 Task-Conditional Regulatory Archetypes

The three-condition results reveal that the GF operates through at least three qualitatively distinct regulatory modes depending on task structure, each with different implications for the coordination-without-collapse balance.

Filter: In the permuted sequential classification task, the active condition drifts toward lower interference relative to the passive baseline while maintaining comparable ER, consistent with the field acting as a filter that suppresses redundant co-activation without degrading representational diversity.

Integrator: The clearest evidence for the GF’s regulatory value emerges in sCIFAR-10. While the passive baseline remains at chance-level performance, the active condition achieves substantial accuracy and ER gains. In this setting of isolated IndRNN units, the GF provides a vital cross-unit signal, acting as an integrator that creates shared temporal context. The active condition’s high-entropy trajectory during loss reduction is consistent with the "fitting phase" of the information bottleneck (Tishby and Zaslavsky, 2015), where task-relevant variance is retained. Conversely, the passive condition’s plateau suggests that cross-unit coordination is a prerequisite for entering this fitting phase.

Disruptive scaffold: In the standard sequential digit recognition task, the active condition shows only modest improvement over the passive baseline, but the representational trajectory is more ambiguous. The active condition maintains elevated entropy throughout training, while Sobol analysis identifies H_{inertia} as the dominant driver of all metrics. The temporal scaffold imposed by the GF introduces coordination overhead that can conflict with local feature learning in a task where such correlations are directly exploitable. A more extreme form of this effect, where coordination degrades performance, is observed in the adding task below.

5.4 The adding task: Appraisal of a failure Mode

The adding task results present the clearest challenge to the coordination-without-collapse claim and require direct engagement. Across both hidden unit configurations, the active condition underperforms the passive baseline, and the probe condition shows degraded validation performance.

Several factors likely contribute. First, the adding task requires precise selective retention: exact numerical values must be preserved across long sequences of noise. The GF imposes shared oscillatory modulation, introducing correlated dynamics that conflict with the need for decoupled, stable units. The Sobol interaction matrix supports this interpretation, showing strong interactions between inertia and both jitter and field strength that increase co-activation, disrupting independent retention. Extreme S_I/S_T divergence confirms disruption arises from higher-order parameter interactions rather than any single factor.

Second, the extreme sensitivity values observed for entropy likely reflect instability in the Sobol estimator due to limited sampling, rather than meaningful effects, and should be interpreted cautiously.

Third, the reversal analysis shows near-zero differences across conditions, indicating all models learn a time-invariant integration heuristic rather than true temporal ordering, consistent with similar activity patterns. Performance differences therefore arise within the same statistical strategy rather than distinct processing modes, and the GF’s temporal scaffold provides no benefit for this task.

6: Conclusion

6.1 Summary of Principal Findings

This study evaluated whether a biologically motivated GF mechanism, implemented as a leaky integrator over population activity that feeds back to modulate neuronal pre-activations, produces measurable effects on representational geometry and task performance across four sequential benchmarks. The three-condition design (passive, probe, active) isolates effects attributable to the field mechanism rather than architectural or observational factors.

Against the two hypotheses stated at the outset, the findings are as follows.

Hypothesis 1 (representational rigidity): Active GF modulation shows partial support. In sCIFAR-10, the active condition uniquely reaches a high-rank specialisation regime without the rank narrowing seen in the passive baseline. In MNIST benchmarks, active and passive trajectories converge, providing no clear confirmation. The formal success criterion (σ ER reduction, $p < 0.05$) cannot be tested due to the single-seed design and remains unresolved. The sCIFAR-10 result provides directional but not inferential support.

Hypothesis 2 (non-pathological coordination): Supported by the Sobol bifurcation analysis. The bifurcation success criterion (interference exceeding 0.7 alongside a corresponding ER decrease) was observed in psMNIST beyond $H_{\text{inertia}} \approx 0.92$, while the segregation criterion (synchrony elevated above the probe baseline while interference remains below 0.4) was met in the sCIFAR-10 active condition at convergence, partially satisfying Hypothesis 2 within task-specific parameter regimes. Below this threshold, the system operates in a regime of moderate synchrony and low interference, consistent with coordination-without-collapse. Formal verification of synchrony thresholds requires per-epoch extraction and remains a numerical gap. Neither criterion was met in the Adding Task, where the active condition operated in an elevated-interference regime throughout, consistent with the disruptive scaffold archetype described in Section 5.3

Performance results are conditional: GF provides the greatest benefit where baseline architectures fail. The 4.6-fold improvement on sCIFAR-10 occurs from a chance-level passive baseline, indicating that isolated channels cannot solve the task without cross-unit coordination. The GF's integrator function closes the cross-channel coordination gap that the architecture otherwise lacks. GF closes this gap. In contrast, on the adding task, global modulation disrupts the independence required for precise regression, leading to worse MSE.

These results define a boundary condition: population-level feedback is beneficial when tasks require cross-unit coordination, but detrimental when independent representations must be maintained over time.

6.2 Theoretical contributions

This study makes three theoretical contributions beyond its empirical results. First, it provides a formal operationalisation of the regulatory gap in sequence modelling: the absence of a mechanism capable of sensing and feeding back population-level representational state during the forward pass, distinct from optimisation-level processes such as Adam or batch normalisation. The GF addresses this gap with a parameter-minimal implementation, requiring no additional learned weights beyond the neuron-gain vector in the heterogeneous condition. Second, the five-metric dissociation framework provides a reusable protocol for distinguishing beneficial compression from pathological redundancy. ER, synchrony, interference, temporal autocorrelation, and entropy together characterise representational state beyond what any single metric captures. Third, the identification of a H_{inertia} bifurcation point, paralleling edge-of-chaos dynamics, motivates an architectural recommendation: the threshold is task-dependent, suggesting dynamic inertia scheduling could exploit both exploratory and stable regimes during training without manual tuning.

6.3 Limitations and their implications for interpretation

The methodological constraints detailed in Section 3.13 apply directly to interpreting these results. Single-seed training means differences below roughly 5 percentage points cannot be separated from initialisation artefacts; the sCIFAR-10 gain is large enough to be considered directionally robust, the sMNIST gain is not. Low Sobol N renders total-order indices directional rather than precise, and consumer hardware constrains scale generalisation.

6.4 Directions for Future Work

The most important immediate next step is multi-seed replication of the sCIFAR-10 and sMNIST three-condition comparisons to determine whether the observed accuracy gaps are robust to initialisation. Following this, the dynamic H_{inertia} scheduling proposal is the highest-priority architectural modification: annealing H_{inertia} from a low initial value toward the task-specific bifurcation threshold over training could allow the network to exploit both the high-entropy exploratory regime and the stable maintenance regime within a single training run, without requiring the current manual search. Finally, applying dimensionality reduction and unit clustering to the Adding Task hidden states would allow direct evaluation of the memory-unit versus context-unit specialisation hypothesis, which the current analysis can only support qualitatively through the heatmap patterns.

6.5 Closing Statement

This study shows that introducing a low-parameter population-level feedback signal produces measurable and distinct effects on representational dynamics in recurrent networks. These effects depend on task structure, emerging most clearly where cross-unit coordination is otherwise absent. Under these conditions, the GF induces a coordination-without-collapse regime consistent with the CTC framework, while exposing failure modes when independent unit dynamics are required. The primary contribution is mechanistic rather than performance-based: explicitly feeding back population-level state alters how representations are formed, stabilised, and degraded during sequence processing. This suggests improving recurrent

architectures may depend less on scale and more on regulatory signals that allow networks to monitor and shape their collective dynamics.

Appendix

Second order interactions adding task H128

S_2	$H_{inertia}$	S	T_p	ξ
λ	-0.247	-0.231	-0.233	-0.230
$H_{inertia}$		-0.241	-0.012	0.059
S			-0.153	-0.145
T_p				-0.121

Table A1: Second order interactions effects on MSE

S_2	$H_{inertia}$	S	T_p	ξ
λ	-1.045	-1.446	-0.903	-0.539
$H_{inertia}$		-0.120	-1.550	-0.717
S			-0.435	0.591
T_p				-0.428

Table A2: Second order interactions effects on ER

S_2	$H_{inertia}$	S	T_p	ξ
λ	0.227	-0.078	0.223	0.300
$H_{inertia}$		1.682	0.389	1.687
S			0.466	0.456
T_p				0.106

Table A3: Second order interactions effects on interference

S_2	H_{inertia}	S	T_p	ξ
λ	-1.557	-5.410	-1.337	-0.675
H_{inertia}		-1.031	-5.583	0.809
S			2.509	3.676
T_p				-0.998

Table A4: Second order interactions effects on entropy

Unified Benchmark Analysis: Metric Trajectories and Temporal Probing

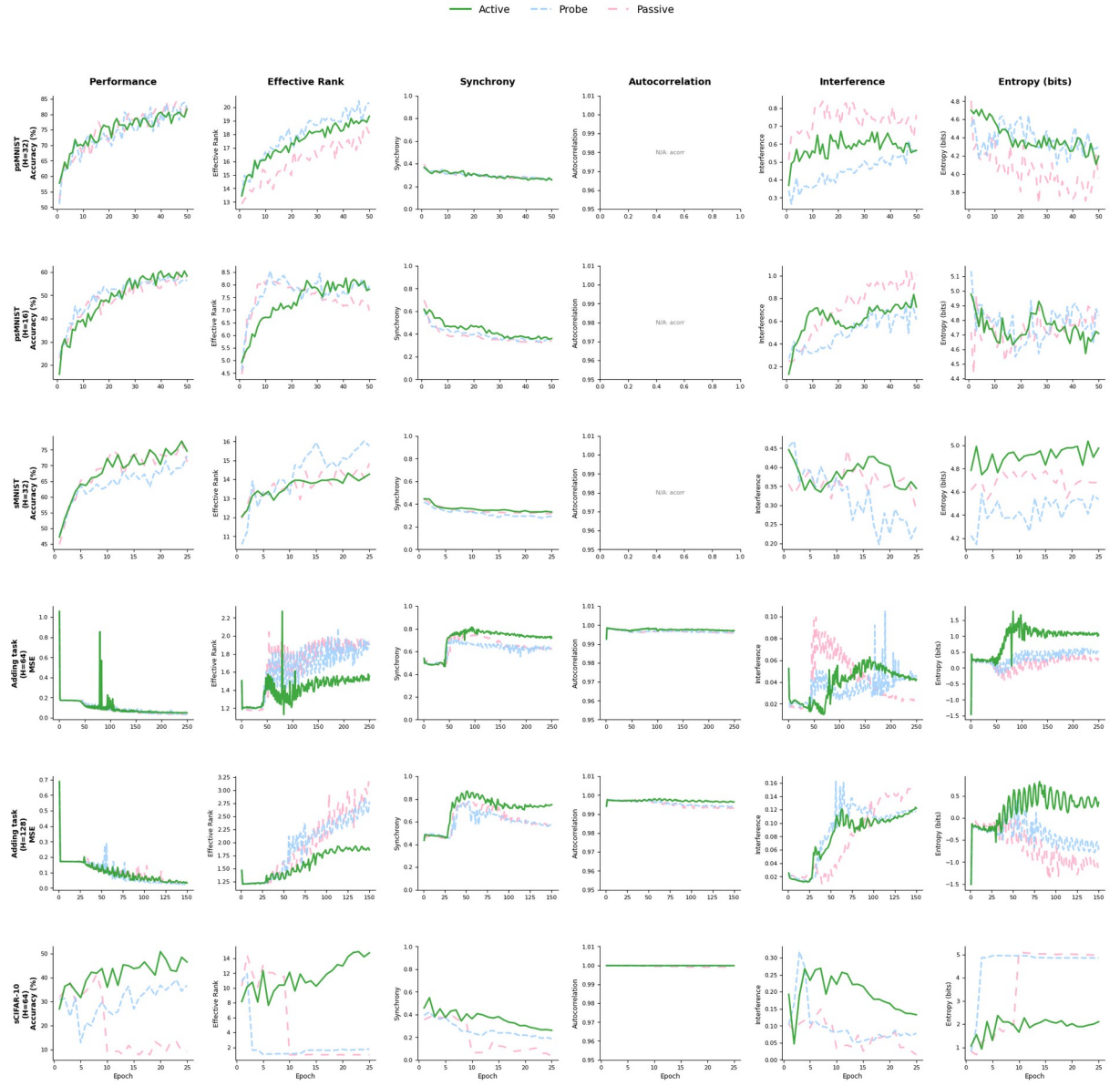


Figure A1

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